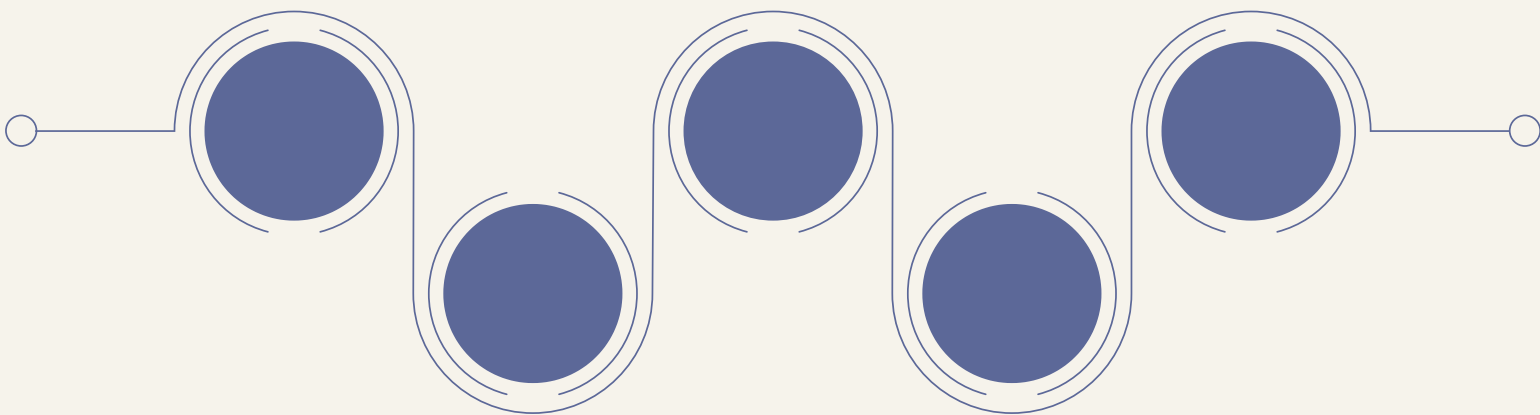
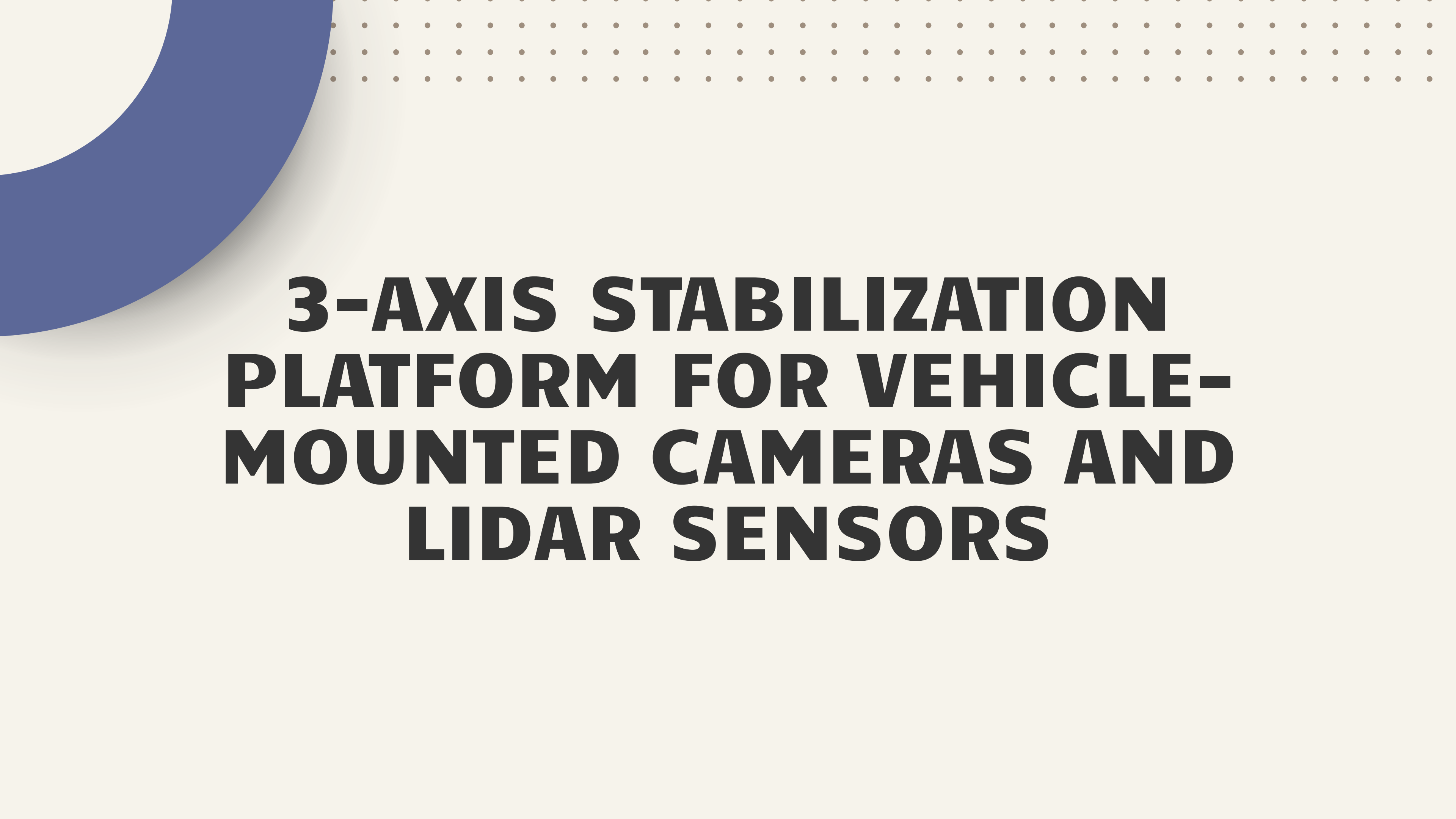




DESIGN LAB-2

PROJECT





3-AXIS STABILIZATION PLATFORM FOR VEHICLE- MOUNTED CAMERAS AND LIDAR SENSORS



GROUP MEMBERS -

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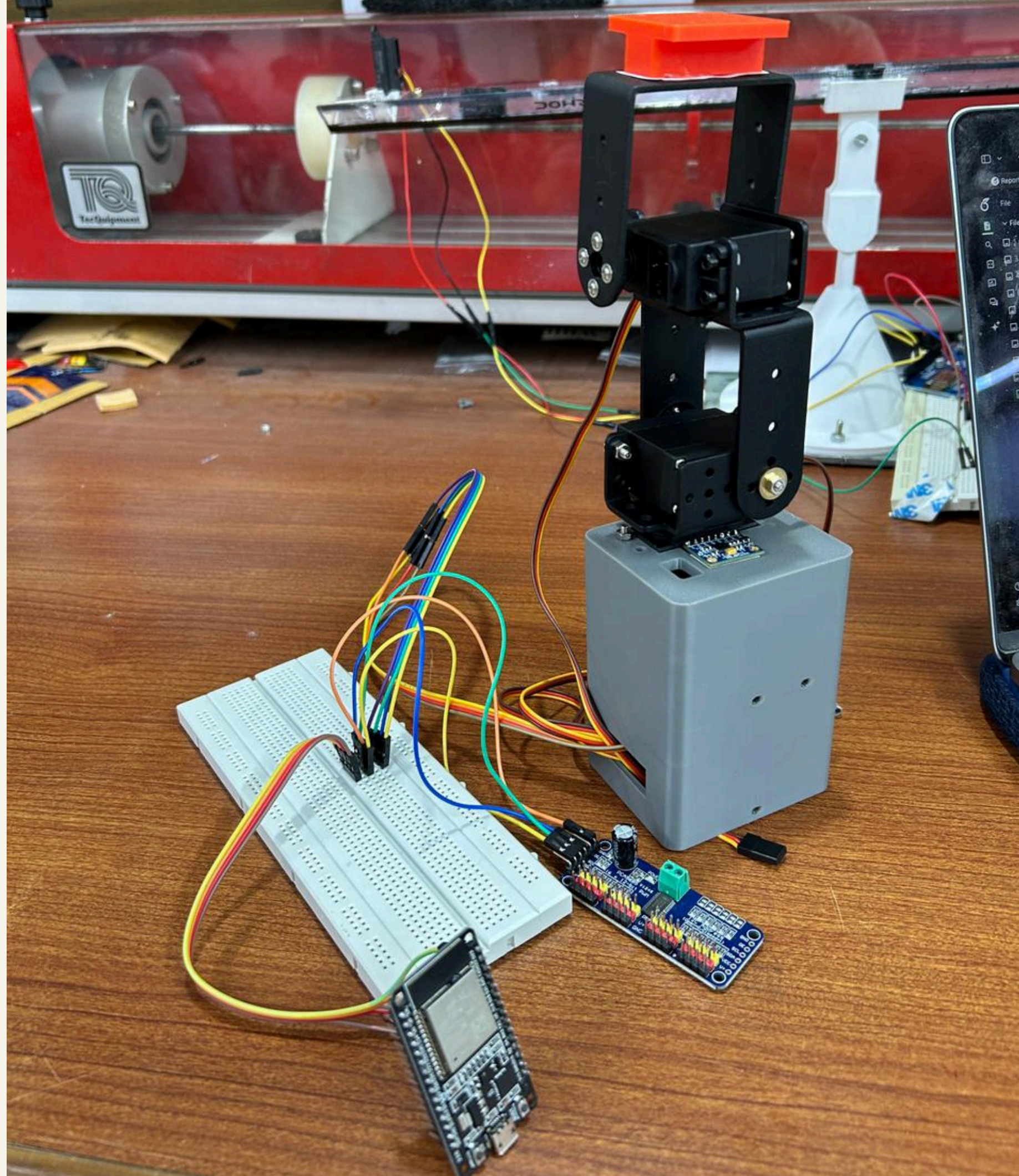
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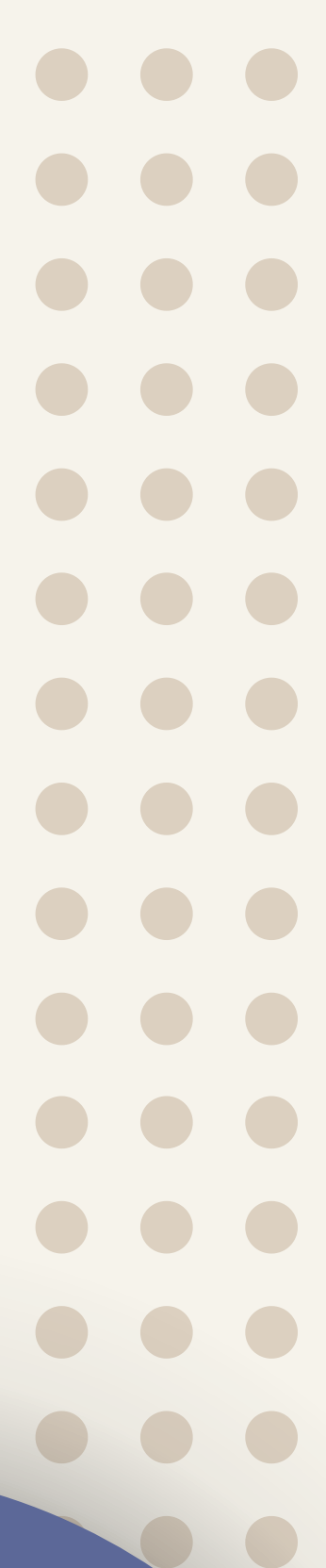
Our Team -







Overview



- **Abstract**

- **Introduction**

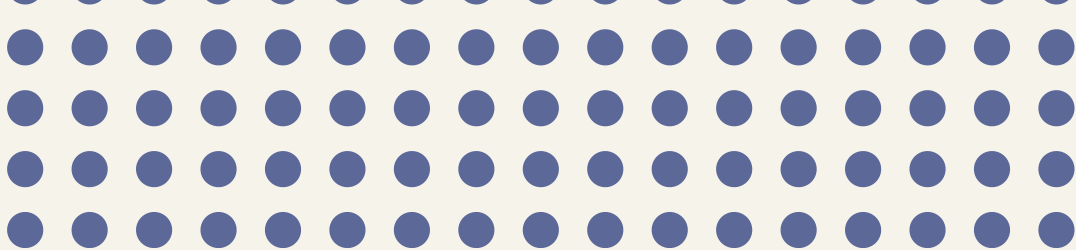
- **Objectives**

- **Methodology**

- **Result**

- **Conclusion**





Abstract

This project presents the design and development of a 3-axis orientation stabilization platform intended for maintaining stable mounting of cameras and LiDAR sensors on vehicles operating in uneven and rough terrain environments. The system addresses the challenge of orientation instability caused by vibrations, sudden accelerations, and multi-directional disturbances that degrade the accuracy of onboard sensing systems in mobile robotics and autonomous platforms.

The proposed system is built around an ESP32 microcontroller as the central processing unit, interfaced with an MPU6050 sensor for real-time acquisition of angular velocity and acceleration data. Orientation estimation is performed for roll, pitch, and yaw axes, where roll and pitch are computed using accelerometer-based trigonometric relations, while yaw is obtained through gyroscope-based angular integration.

A closed-loop Proportional–Integral–Derivative (PID) control algorithm is implemented to continuously minimize the deviation between desired and measured orientation. The control output is mapped to three MG995 high-torque servos through a PCA9685 PWM driver module, enabling real-time mechanical correction of the platform's orientation. The system is designed using a single-platform servo linkage mechanism, allowing coordinated multi-axis stabilization.

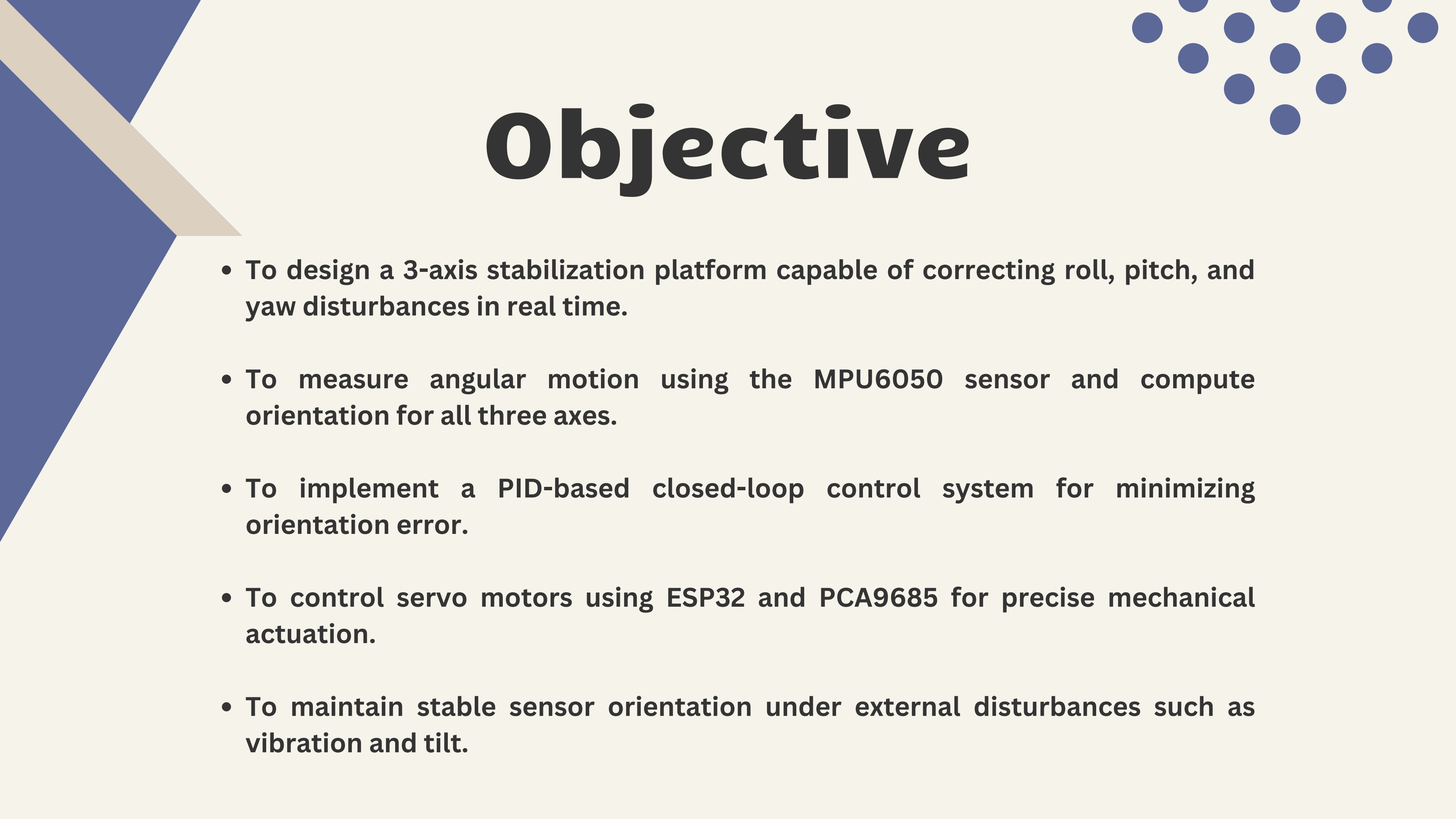




Introduction

In modern surveillance, autonomous navigation, defense, and terrain-mapping systems, maintaining stable sensor orientation is critical for ensuring accurate data collection and operational efficiency. Vehicle-mounted cameras and LiDAR sensors operating on rough or uneven terrain are constantly exposed to vibrations, shocks, tilts, and unpredictable motion disturbances. These external factors can significantly degrade image quality, reduce sensor accuracy, and compromise navigation or monitoring performance.

This project focuses on the development of a servo-actuated 3-axis stabilization platform specifically intended for vehicle-mounted cameras and LiDAR sensors in rough terrain applications. By integrating embedded control systems, real-time motion sensing, and multi-axis mechanical correction, the platform is designed to preserve sensor alignment and enhance data reliability under dynamic conditions.



Objective

- To design a 3-axis stabilization platform capable of correcting roll, pitch, and yaw disturbances in real time.
- To measure angular motion using the MPU6050 sensor and compute orientation for all three axes.
- To implement a PID-based closed-loop control system for minimizing orientation error.
- To control servo motors using ESP32 and PCA9685 for precise mechanical actuation.
- To maintain stable sensor orientation under external disturbances such as vibration and tilt.



Methodology

- Our system implements a closed-loop 3-axis orientation stabilization platform using real-time inertial sensing and servo-based actuation.
- The control pipeline follows:
Sensing → State Estimation → Error Computation → PID Control → Actuation → Feedback



1. Data Acquisition



- The MPU6050 provides two types of measurements -
 1. Accelerometer → measures linear acceleration (a_x , a_y , a_z)
 2. Gyroscope → measures angular velocity (ω_x , ω_y , ω_z)
 - These values are read continuously by the ESP32.
- 
- 

2. Orientation Estimation

We calculate orientation in terms of Roll, Pitch, and Yaw angles -

1. Roll Angle (ϕ) -

$$\phi = \tan^{-1} \left(\frac{a_y}{a_z} \right)$$

2. Pitch Angle (θ) -

$$\theta = \tan^{-1} \left(\frac{-a_x}{\sqrt{a_y^2 + a_z^2}} \right)$$

3. Yaw Angle (ψ) - Yaw cannot be obtained from the accelerometer because rotation around the vertical axis does not affect gravity. So we compute yaw using gyroscope integration -

$$\psi(t) = \psi(t - 1) + \omega_z \cdot dt$$

3. Error Computation

For each axis -

$$e(t) = \theta_{desired} - \theta_{measured}$$

- Roll error -

$$e_{\phi} = 0 - \phi$$

- Pitch error -

$$e_{\theta} = 0 - \theta$$

- Yaw error -

$$e_{\psi} = 0 - \psi$$

4. PID Control

The controller generates a correction signal -

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$$

Discrete Implementation (Used in our ESP32) -

$$u[k] = K_p e[k] + K_i \sum e[k] \cdot dt + K_d \frac{e[k] - e[k - 1]}{dt}$$

5. MATLAB Simulation

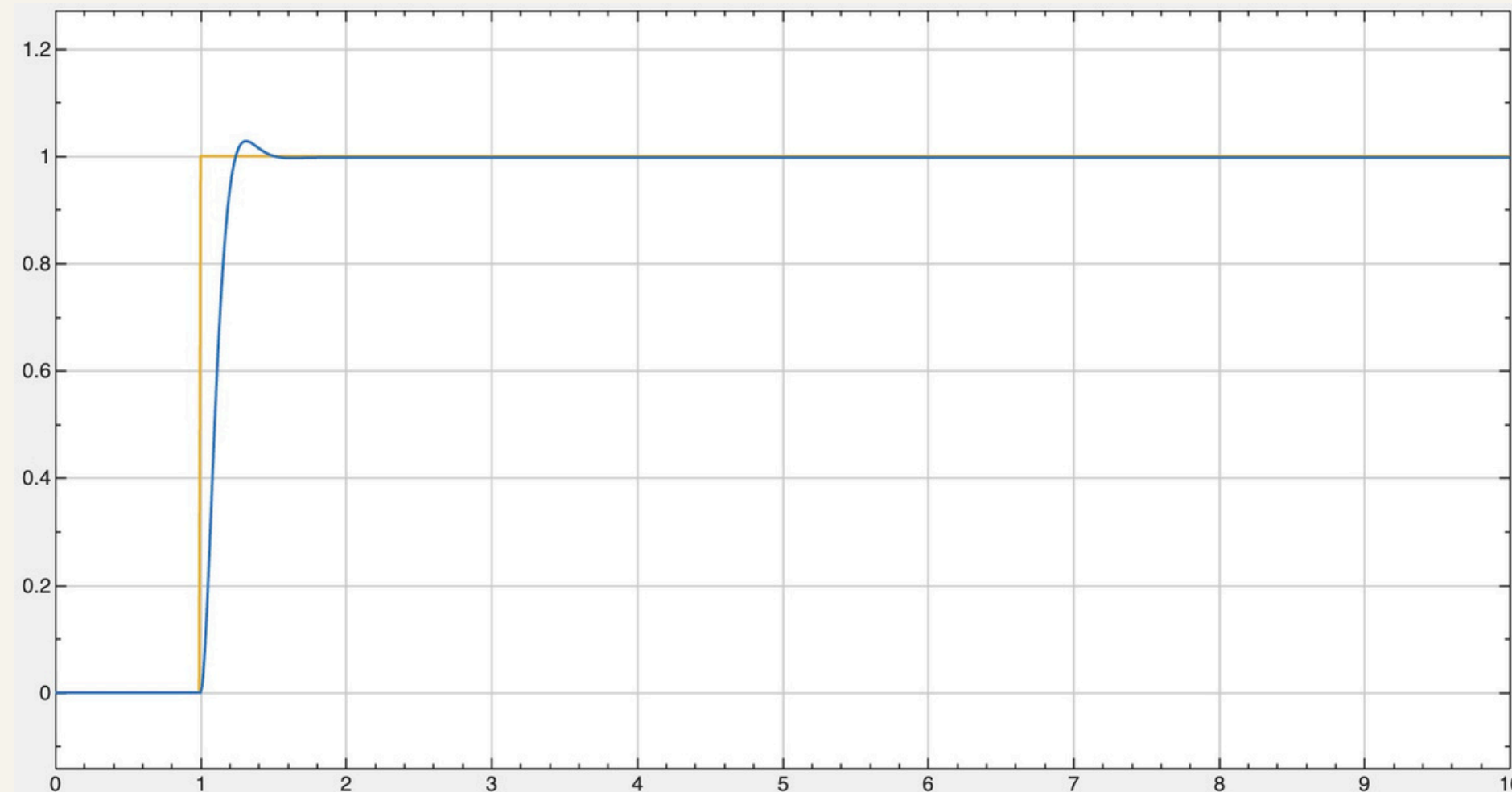
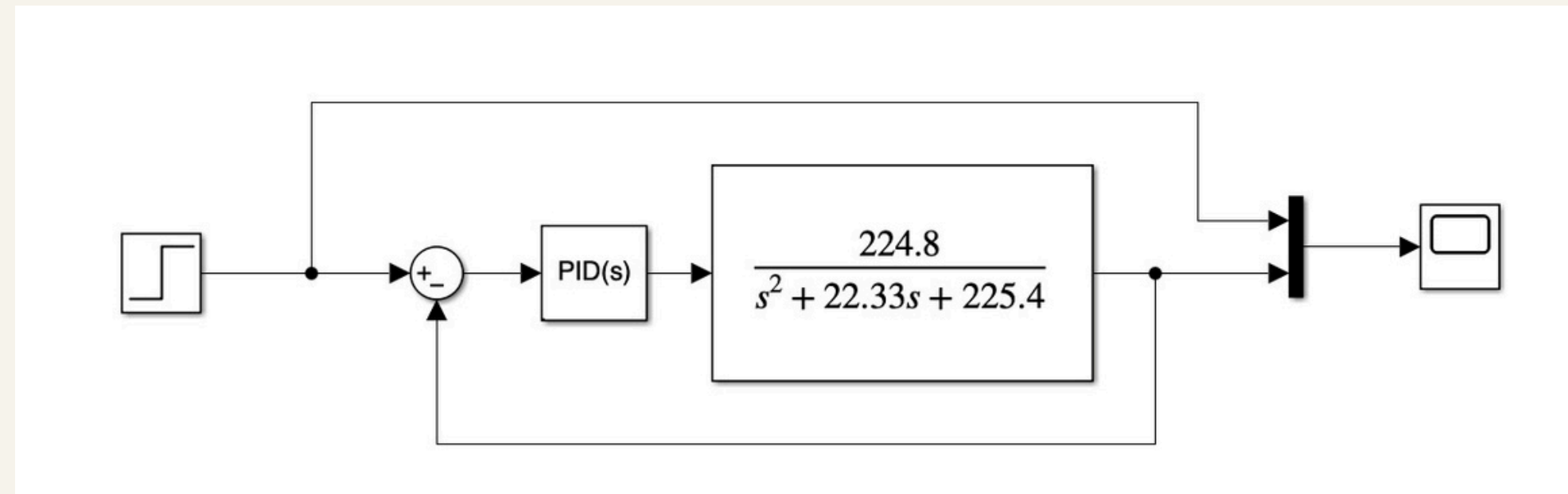
The estimated transfer function of MG995 servo motor is -

$$\frac{\Theta_l(s)}{\Theta_r(s)} = \frac{224.8}{s^2 + 22.33s + 225.4}$$

Reference :

https://www.researchgate.net/publication/357078983_Physical_Modeling_and_Parameters_Identification_of_the_MG995_Servomotor

The values of the PID controller gains K_p , K_i , and K_d were determined using a trial-and-error tuning method based on analysis of the system's step response.

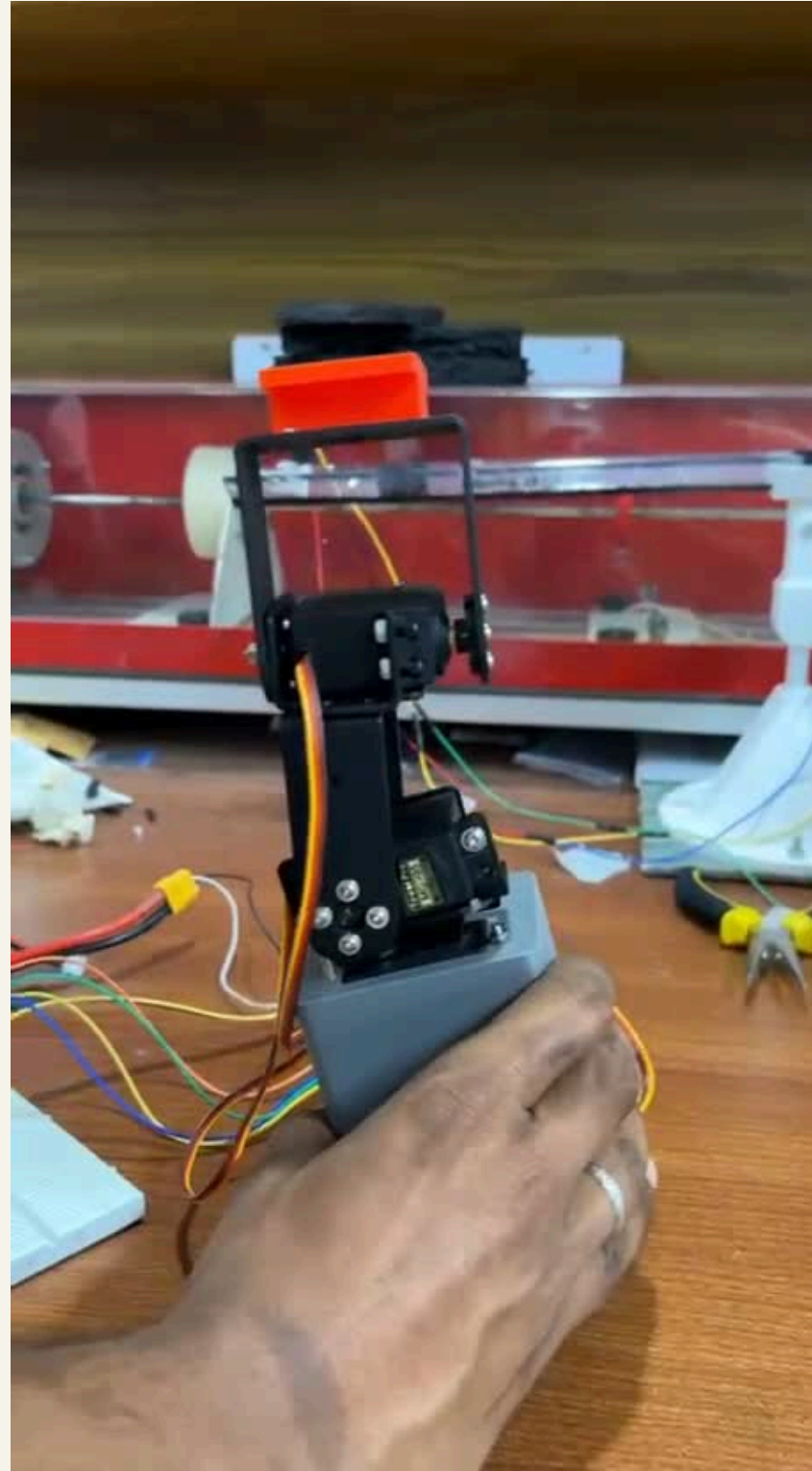


Proportional (P): 20

Integral (I): 2

Derivative (D): 0.2

Result





Conclusion

The project successfully demonstrates the design and implementation of a 3-axis orientation stabilization platform using embedded control and mechatronic principles. By utilizing real-time data from the MPU6050 sensor and applying a PID-based closed-loop control system, the platform effectively reduces disturbances in roll, pitch, and yaw, maintaining stable orientation under external motion.

The system shows reliable performance in terms of fast response, minimal overshoot, and low steady-state error, validating the effectiveness of the control strategy and actuator design. MATLAB-based modeling and simulation further support the practical feasibility of the system.

Overall, the project highlights the integration of control systems, embedded programming, and mechanical design to solve real-world stabilization problems, with strong potential for applications in robotics, autonomous systems, and sensor platforms operating in dynamic environments.



THANK YOU

